DT081A — Laboratory 2 Report

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1 Introduction

This lab studies signals in the time and frequency domains. The tasks include windowing, LTI systems, digital filters and voice analysis.

2 Experimental Setup

I used Python with NumPy, SciPy and Matplotlib. For the first task, I sampled three cosine waves at 250 Hz for 0.6 seconds. Their frequencies were 50, 55 and 120 Hz.

3 Results and Analysis

3.1 Time-Domain Analysis

The signal contains three cosine waves. The FFT shows peaks at 50, 55 and 120 Hz (Figure 1).

Q1. Why does the amplitude spectrum contain negative frequencies?

Answer: The signal contains real cosine waves, so its FFT has matching positive and negative frequencies. The negative side is a mirror of the positive side. It does not mean extra frequencies were added to the signal (Figure 1).

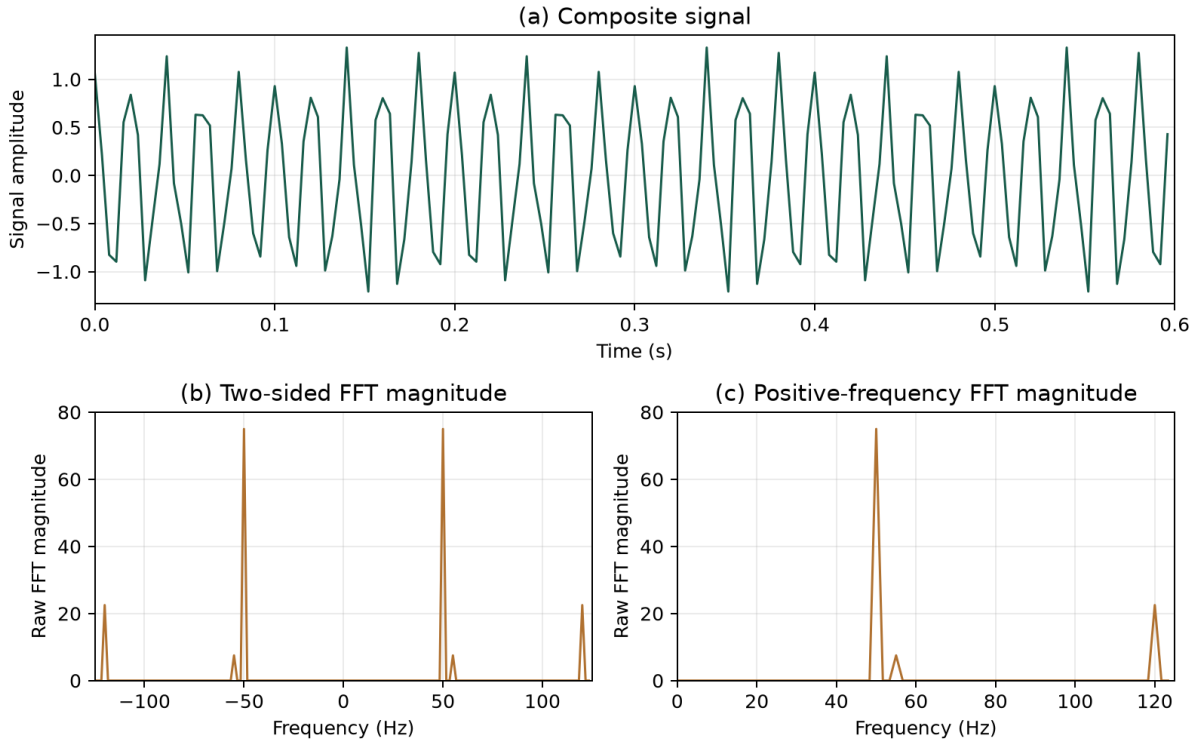


Figure 1: Composite signal, two-sided FFT magnitude and positive-frequency FFT magnitude. FFT magnitudes are unnormalized.

4 Discussion

The FFT showed the same frequencies that were used to make the signal. The positive and negative frequency peaks matched because the signal was real.

5 Conclusion

The first experiment showed how the FFT can identify the frequencies in a signal.